Dr Rowan Lindsay-Smith

Research Software Engineering & Computational Linguistics

University of Oxford

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Research Software Engineer and technical lead specialising in scientific software and machine learning infrastructure. Technical lead across UKRI- and ERC-funded research programmes worth over £12M, designing, developing, deploying and maintaining production-quality research software for multilingual speech recognition, experimental research and large-scale linguistic fieldwork.

University of Oxford

Postdoctoral Researcher

December 2021 — Present

Technical Leadership

- Technical lead for software development across the **£2.06M UKRI Pertinacity** and **€10.09M ERC Synergy PAAL** programmes.
- Coordinated software development between academic researchers and the FlexSR commercial spin-out, planning technical milestones and software delivery.
- Co-authored approximately **70%** of the successful **€150k ERC Proof of Concept FLEX-CODESWITCH** proposal, leading the technical work packages.
- Supervised research assistants developing research software and conducting large-scale speech annotation, supported by custom automation that streamlined annotation workflows.

Major Software Systems

FlexSR

Multilingual speech-recognition platform

Technical lead for a multilingual speech-recognition platform supporting academic research, language documentation, and commercialisation.

- Lead development of desktop, web and machine learning applications
- Design speech-processing pipelines, model training and evaluation infrastructure.
- Develop machine learning workflows for automated transcript verification
- Underpins research and commercial development across the UKRI Pertinacity, ERC PAAL and ERC Proof of Concept FLEX-CODESWITCH programmes.
- Improved application runtime performance by 40%
- Built React demonstration platform supporting research dissemination and commercial partners.

XMod

Psycholinguistic experimentation platform

Architected and developed a cross-platform Java Swing behavioural experimentation platform replacing legacy Visual Basic software.

- Preserve backwards compatibility with existing experiment definitions and bespoke laboratory hardware
- Modernise the codebase with an MVC architecture and comprehensive JUnit testing.
- Supports monomodal and crossmodal psycholinguistic experiments involving up to 16 simultaneous participants
- Remains under active development as the laboratory’s primary platform for reaction-time experimentation.

Harken

Speech data collection platform

Designed and deployed a full-stack configurable React-based speech data collection platform. Enabled independent linguistic fieldwork in India, collecting 4,800 speech recordings across three experiments for automated preparation and annotation through AudioPipeline.

Additional Software Projects

AudioPipeline

Configurable pipelines for automated speech corpus preparation and annotation using Montreal Forced Aligner, substantially reducing manual annotation effort while supporting scalable research assistant workflows. Comprehensive pytest unit testing supports reliable long-term use.

SpliceHelper

Python application automating configuration of more than 10 psycholinguistic experiments for the XMod experimentation platform, replacing manual preparation while maintaining compatibility with existing laboratory workflows.

Conference Website

Developed and deployed a responsive React website for a five-day international conference (400 attendees), replacing a static site with a maintainable component-based architecture that supported rapid content updates during planning and delivery.

Technical Stack

Languages	Python · Java · JavaScript · Bash
Scientific Computing	PyTorch · scikit-learn · NumPy · pandas
Testing	pytest · JUnit
Engineering	Software Architecture · Automated Testing · CI/CD · Deployment · Technical Leadership
Frameworks	FastAPI · React · Java Swing
Developer Tools	Git · GitHub · GitLab · Bitbucket · Linux · HPC · Montreal Forced Aligner · Praat

Previous Academic Appointments

Postdoctoral Research Fellow, University of Surrey	2021 — 2022
Departmental Lecturer in Phonology, University of Oxford	2020 — 2021
Associate Lecturer, Oxford Brookes University	2020 — 2022
Outreach Officer, University of Oxford	2021

Education

University of Oxford	Oxford, UK
DPhil, Comparative Philology & General Linguistics	2021
MPhil, General Linguistics & Comparative Philology (Distinction)	2018
BA, Oriental Studies (Arabic with Turkish) (1st)	2016

Selected Publications

Reetz, H. and Lindsay-Smith, E. (forthcoming). *Phonology–Phonetics Interface in Automatic Speech Recognition*. Festschrift Volume.

Kennard, H., Lindsay-Smith, E., Lahiri, A. & Maiden, M. (Eds.) (2025). *Historical Linguistics 2022*. John Benjamins Publishing Company.

Lindsay-Smith, Emily (2024). ”Affix not clitic-based vowel shortening in Modern Arabic Varieties”. *Transactions of the Philological Society*.

Lindsay-Smith, E. et al. (2024). ”Analogy in Inflection”. *Annual Review of Linguistics*.

Awards

R. H. Robins Prize — Philological Society of Great Britain	2021